

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12417141
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Marson; Mark Evan et al.

Generating a target data based on a function associated with a physical variation of a device

Abstract

A value corresponding to a physical variation of a device may be received. Furthermore, helper data associated with the physical variation of the device may be received. A result data may be generated based on a combination of the value corresponding to the physical variation of the device and the helper data. An error correction operation may be performed on the result data to identify one or more code words associated with the error correction operation. Subsequently, a target data may be generated based on the one or more code words.

Inventors: Marson; Mark Evan (Carlsbad, CA), Best; Scott C. (Palo Alto, CA), Handschuh; Helena (Miami, FL), Wu; Winthrop John (Pleasanton, CA)

Applicant: Cryptography Research, Inc. (San Jose, CA)

Family ID: 1000008818806

Assignee: Cryptography Research, Inc. (San Jose, CA)

Appl. No.: 18/644084

Filed: April 23, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240345916 A1	Oct. 17, 2024

Related U.S. Application Data

continuation parent-doc US 16972560 US 12013751 WO PCT/US2019/035627 20190605 child-doc US 18644084
us-provisional-application US 62683483 20180611

Publication Classification

Int. Cl.: G06F11/10 (20060101); H04L9/08 (20060101); H04L9/32 (20060101)

U.S. Cl.:

CPC G06F11/10 (20130101); H04L9/0866 (20130101); H04L9/3278 (20130101);

Field of Classification Search

CPC: G06F (11/10); G06F (11/1048); H04L (9/0866); H04L (9/3278); H04L (2209/34)

USPC: 714/752

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8610454	12/2012	Plusquellic et al.	N/A	N/A
8983067	12/2014	Murray	N/A	N/A
2012/0072737	12/2011	Schrijen et al.	N/A	N/A
2012/0183135	12/2011	Paral	380/44	G09C 1/00
2013/0051552	12/2012	Handschuh et al.	N/A	N/A
2014/0218067	12/2013	Li	326/8	H04L 9/3278
2014/0325237	12/2013	Van Der Leest et al.	N/A	N/A
2015/0026545	12/2014	Yu	714/780	H03M 13/134
2015/0234751	12/2014	Van Der Sluis et al.	N/A	N/A
2016/0156476	12/2015	Lee	380/44	H04L 9/0866
2016/0227348	12/2015	Guo et al.	N/A	N/A
2017/0155389	12/2016	Zhao et al.	N/A	N/A
2017/0310489	12/2016	Van Der Sluis	N/A	H04L 9/3278
2018/0219669	12/2017	Chen et al.	N/A	N/A
2019/0147967	12/2018	Tuyls	365/96	G11C 17/18

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2012-519987	12/2011	JP	N/A
WO-2016-058793	12/2015	WO	N/A
WO-2016-102164	12/2015	WO	N/A

OTHER PUBLICATIONS

EP Extended European Search Report with Mail Date Feb. 24, 2022 re: EP Appln. No. 19819599.2.
21 pages. cited by applicant

EP Response filed on Sep. 21, 2022 in Response to the Extended European Search Report Dated

Feb. 24, 2022 and the Official Communication Pursuant to Rules 70(2) and 70a(2) EPC dated Mar. 15, 2022 for EP Appln. No. 19819599.2. 27 pages. cited by applicant
JP Office Action mailed Mar. 29, 2023 re: JP Appln. No. 2020-544757. 7 pages (w/translation). cited by applicant
Merli, Dominik et al, "Protecting PUF Error Correction by Codeword Masking", IACR Cryptology ePrint Archive, pp. 334-349, May 2013. [retrieved on Aug. 2, 2019]. Retrieved from the Internet: <URL: <https://pdfs.semanticscholar.org/a037/c8de695e6371a8c0ce1defb3197b7090f089.pdf>>; entire document. 16 Pages. cited by applicant
Notification of Transmittal of the International Search Report and the Written Opinion of the International Searching Authority, or the Declaration with Mail Date Aug. 29, 2019 re: Int'l Appln. No. PCT/US19/035627. 38 pages. cited by applicant

Primary Examiner: Kerveros; James C

Attorney, Agent or Firm: Lowenstein Sandler LLP

Background/Summary

RELATED APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 16/972,560, Dec. 4, 2020, which is a National Phase Application of International Application No. PCT/US2019/035627, filed Jun. 5, 2019, which claims the benefit of U.S. Provisional Application No. 63/683,483, filed Jun. 11, 2018, the entire contents of all are incorporated by reference.

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure will be understood more fully from the detailed description given below and from the accompanying drawings of various implementations of the disclosure.
 - (2) FIG. 1 illustrates an example device including a physically unclonable function in accordance with some embodiments of the present disclosure.
 - (3) FIG. 2 is a flow diagram of an example method to generate helper data and a target data by using a physically unclonable function in accordance with some embodiments.
 - (4) FIG. 3A is a flow diagram of an example method to generate helper data for a physically unclonable function in accordance with some embodiments of the present disclosure.
 - (5) FIG. 3B illustrates the generation of helper data in accordance with some embodiments of the present disclosure.
 - (6) FIG. 4A is a flow diagram of an example method to generate a target data by using a physically unclonable function in accordance with some embodiments of the present disclosure.
 - (7) FIG. 4B illustrates the generation of the target data in accordance with some embodiments of the present disclosure.
 - (8) FIG. 5 illustrates a block diagram of an embodiment of a computer system in which some embodiments of the disclosure may operate.
-

Description

DETAILED DESCRIPTION

- (1) Aspects of the present disclosure are directed to generating a target data based on a function associated with a physical variation of a device. In general, a semiconductor device may include physical variations as a result of the semiconductor fabrication process of the semiconductor

device. The physical variations may be unique to a particular semiconductor device. For example, the physical variation may be a unique hardware characteristic or behavior of the semiconductor device. The physical variation may be randomly introduced to the semiconductor device as part of the semiconductor fabrication process. Such a physical variation in a semiconductor device may be associated with a physically unclonable function (PUF) that generates a value that is representative of the physical variation.

(2) The physically unclonable function may be used to generate a target data that is used to perform operations at a device. For example, the PUF may correspond to an unpredictable or unknown but constant value for a particular semiconductor device. The value corresponding to the PUF may be used to generate a target data such as a secret cryptographic key to be used by the device. Such a cryptographic key may be common among multiple other devices. Since the value corresponding to the different PUFs in different devices may be unknown and unpredictable, additional data (i.e., helper data) may be used to facilitate the generation of the expected target data when using the value corresponding to the PUF for each of the devices. For example, the helper data may be used to determine which values corresponding to the PUF should be selected and used to generate the target data. As a result, the combination of the corresponding helper data and the value corresponding to the PUF may be used to generate the same target data for each of the devices when the value corresponding to the PUF may be different between the devices.

(3) As an example, the PUF may correspond to physical variation in a static random access memory (SRAM) where particular bits of the SRAM may be biased towards a value of '0' or a value of '1' at an initialization of a device that includes the SRAM. The helper data may identify which particular biased bits of the SRAM should be selected and used as the value corresponding to the PUF that are used to generate the target data. The helper data may be generated by the PUF after the values corresponding to the PUF are selected and the helper data may be stored in a memory. Subsequently, the helper data may be retrieved from the memory and provided to the PUF at a later time so that the corresponding values of the PUF may be selected to generate the same target data. As a result, the helper data may be used to select particular bits of a repeatable but random value of the PUF to generate the target data.

(4) The use of the value corresponding to the PUF (i.e., the PUF value) may result in an exporting of the PUF value during the manufacturing of a device that includes the PUF. Furthermore, the use of the value corresponding to the PUF may be used in an additional operation (e.g., to encrypt a cryptographic key with the value corresponding to the PUF). However, such exporting of the PUF value during the manufacturing of the device or the use of the PUF value in an additional operation may expose the PUF value so that an unauthorized entity may retrieve the PUF value and compromise the security of the device. For example, if the PUF value is used to generate a cryptographic key to be used by the device, then the cryptographic key may be retrieved or generated by the unauthorized entity after the PUF value has been exported.

(5) Aspects of the present disclosure address the above and other deficiencies by utilizing a PUF corresponding to physical variations in, for example (but not limited to), SRAM. For example, code words used in an SRAM error correction operation may be generated based on values of a target data and random numbers. In general, the helper data may further be a combination of the target data and the code words that are based on the values of the target data and the random numbers. Subsequently, the helper data may be used to generate the target data. For example, the output of the PUF may be received and combined with the helper data. An error correction operation may be performed on the combination of the output of the PUF and the helper data to generate code words. Subsequently, the code words may be used to determine the corresponding target data.

(6) Advantages of the present disclosure include, but are not limited to, an improved security in the target data. For example, since the value of the PUF is not exported and thus less likely to be retrieved by an unauthorized entity, the target data that is generated based on the value of the PUF

may be considered to be more secure than when the value of the PUF is exported. Thus, the operations of a device that uses the PUF to generate a target data corresponding to a cryptographic key when performing the operations may be more secure. Furthermore, a device manufactured for a first customer may be programmed at manufacture to generate a first target value while devices manufactured for a second customer may be programmed to generate a second target value.

(7) FIG. 1 illustrates an example device **100** including a physically unclonable function. The device **100** may include an integrated circuit where a portion of the integrated circuit is associated with a physical variation corresponding to the physically unclonable function that is used to generate a target data. In general, the physically unclonable function may be a circuit that corresponds to, but is not limited to, a static random access memory (SRAM) PUFs (which rely on random voltage offsets), timing PUFs (which rely on unpredictable race conditions), impedance PUFs (which rely on resistance, capacitance or inductance mismatch), etc. Such PUF circuits may utilize helper data that is stored in a non-volatile memory to achieve long-term stability of the output of the PUF circuit (e.g., the PUF value).

(8) As shown in FIG. 1, the device **100** may include the physically unclonable function **110**, a memory **120**, and device components **130**. The memory **120** may be a non-volatile memory of the device and the physically unclonable function **110** may correspond to an integrated circuit or a structure or portion of an integrated circuit. Furthermore, the device components **130** may be circuitry that performs functionality associated with a cryptographic operation. For example, the cryptographic operation may be to encrypt and/or decrypt input data by using a target data that is generated by using the physically unclonable function **110**.

(9) The physically unclonable function **110** may generate a subsequent target data **141** based on a physical variation of the PUF **110** and helper data **140** from the memory **120**. For example, the physically unclonable function **110** may receive a target data **150**. In some embodiments, the target data **150** may be an intended cryptographic key for a class of devices. For example, the target data **150** may be a cryptographic class key that is intended to be used or generated by each device (e.g., the device **100** and other devices) in the class of devices. After receiving the target data **150**, the physically unclonable function **110** may generate helper data **140** based on the target data **150** and values corresponding to the physical variation associated with the physically unclonable function **110**. In some embodiments, the physical variation of the physically unclonable function **110** may correspond to multiple values. For example, a first value may correspond to a first physical variation in the structure of the physically unclonable function **110** and a second value may correspond to a second physical variation in the structure of the physically unclonable function **110**. In some embodiments, if the physically unclonable function **110** corresponds to an SRAM, then the first physical variation may correspond to whether a first bit of the SRAM is more biased towards a value of '0' or a value of '1' at initialization or startup of the device **100** and the second physical variation may correspond to whether a second bit of the SRAM is more biased towards a value of '0' or a value of '1' at initialization or startup of the device **100**. Thus, different physical variations in the structure of the physically unclonable function **110** may each correspond to a different bit that may be used to generate the target data **150**. The helper data **140** may identify which particular bits corresponding to the different physical variations of the physically unclonable function **110** should be selected and used to generate the received target data **150** or may be used to generate the received target data **150**.

(10) As an example, the target data **150** may be a value of '00110.' The physically unclonable function **110** may provide values '001111111110' as a result of its different physical variations. The helper data **140** may identify that the first four bits '0011' and the last bit '0' should be selected from the physically unclonable function **110** to generate the target data **150**. Since the helper data includes information corresponding to a utilization of certain bits or other such information, then the information from the helper data alone does not reveal the desired target value to an attacker or other such unauthorized entity that is able to obtain the helper data. In the same or alternative

embodiments, the helper data may include additional information, such as the first four bits may be selected without being modified and then the fifth bit may be selected and inverted or performed with another type of modification. In some embodiments, if the physically unclonable function **110** corresponds to an SRAM, then the helper data **140** may correspond to code words and the output of the physically unclonable function **110** as described in further detail below.

(11) Referring to FIG. **1**, the helper data **140** may be stored in the memory **120** after being generated by the physically unclonable function **110**. Subsequently, the device components **130** may request that the target data **150** be provided. In response, the helper data **140** may be provided to the physically unclonable function **110** to generate the subsequent target data **141** that may match the original target data **150** that was used to generate the helper data **140**. In some embodiments, the subsequent target data **141** may correspond to a cryptographic key to be used by the device components **130** or a value that is used to generate the cryptographic key.

(12) FIG. **2** is a flow diagram of an example method **200** to generate helper data and a target data by using a physically unclonable function. The method **200** may be performed by processing logic that may include hardware (e.g., processing device, circuitry, dedicated logic, programmable logic, microcode, hardware of a device, integrated circuit, etc.), software (e.g., instructions run or executed on a processing device), or a combination thereof. In some embodiments, the method **200** may be performed by the physically unclonable function **110** and/or other device components **130** of the device **100** of FIG. **1**.

(13) As shown in FIG. **2**, the method **200** may begin, at block **210**, with the processing logic providing a target data to a physically unclonable function (PUF). For example, a cryptographic key that is intended to be replicated at a later time may be provided to the PUF. The processing logic may further generate helper data from the PUF based on the target data (block **220**). For example, the helper data may provide additional data to reproduce the target data by using the values from the PUF and/or identify particular bits corresponding to the physical variations of the PUF that should be selected and used to replicate the target data as an output value of the PUF. In some embodiments, the PUF may correspond to an SRAM and the helper data may be a combination of an output value of the PUF and code words. Further details with regards to the generation of the helper data are described in conjunction with FIGS. **3A** and **3B**. Subsequently, the processing logic may store the helper data at a memory (block **230**). The helper data may be stored at the memory for subsequent retrieval to generate the target data at a later time. The processing logic may further receive a request to generate the target data with the PUF (block **240**). For example, the request to generate the target data may be received after the helper data has been generated and stored at the memory. The request to generate the same target data may be in response to a request from a device component for a cryptographic key corresponding to the target data that is to be used in an operation by the device component. The processing logic may provide the helper data at the memory to the PUF (block **250**). For example, the helper data that was previously generated by the PUF based on the previously received target data may be provided to the PUF at a later time. The processing logic may subsequently generate the target data based on the PUF and the helper data (block **260**). For example, the helper data may be used to generate or select values corresponding to the PUF that reflect the previous target data that was received by the PUF and used to generate the helper data. In some embodiments, the helper data may be combined with the output of the PUF to produce a result where error correction is performed on the result to retrieve code words. Furthermore, the code words may be used to derive or generate the target data. Further details with regards to generating the target data from the code words are described in conjunction with FIGS. **4A** and **4B**. The code words may be error correcting codes that are used in an error correction operation.

(14) As such, physical variations of the circuitry of the device (e.g., the PUF) may represent multiple bits or values. The helper data may be used to identify a subset of the bits or values represented by the physical variations and/or additional data (e.g., code words) that are used in

conjunction with the bits or values represented by the physical variations to generate the target data. For example, the helper data may identify a particular number and an ordering of the bits represented by the physical variations and code words to be used with an error correction operation so that the number and ordering corresponds to or is the target data after the performing of the error correction operation.

(15) FIG. 3A is a flow diagram of an example method **300** to generate helper data for a physically unclonable function. The method **300** may be performed by processing logic that may include hardware (e.g., processing device, circuitry, dedicated logic, programmable logic, microcode, hardware of a device, integrated circuit, etc.), software (e.g., instructions run or executed on a processing device), or a combination thereof. In some embodiments, the method **300** may be performed by the physically unclonable function **110** and/or other device components **130** of the device **100** of FIG. 1.

(16) The method **300** may correspond to the generation of helper data for a PUF that corresponds to an SRAM. As shown in FIG. 3A, the method **300** may begin, at block **310**, with the processing logic receiving a target data for a physically unclonable function that corresponds, in this example, to a static random access memory (SRAM). In other embodiments, the PUF may correspond to any other type of PUF circuit that utilizes helper data to ensure stability (for example, but not limited to, voltage-offset-based PUFs, timing-based PUFs or impedance-based PUFs). The processing logic may receive random values from a random number generator (block **320**). For example, a device that includes the PUF may further include a random number generator component that generates one or more random values or numbers. The processing logic may subsequently generate code words associated with error correction based on the random values and the target data (block **330**). The generated code words may be selected from a code word basis that includes a set of elements (e.g., code words) to be used in an error correction operation. A first portion of the code word basis may be selected based on values of the bits of the target data and a second portion of the code word basis may be selected based on values of the bits of the random values. The code word basis may include a first subset that corresponds to or is equal to a number of bits in the target data and a second subset that corresponds to or is equal to a number of random values that are generated. A particular code word may be selected when a value of a corresponding bit is a '1' and the particular code word may not be selected when the value of the corresponding bit is a '0.' Thus, code words from the first subset may be selected based on corresponding bit values of the target data being a '1' and code words from the second subset may be selected based on corresponding bit values of the random numbers being a '1.'

(17) Referring to FIG. 3A, the processing logic may further perform an exclusive-or (XOR) operation between an output value of the PUF and the code words to generate helper data for the PUF corresponding to the SRAM (block **340**). For example, an output value of the PUF may be combined with the generated code words to generate the helper data. Subsequently, the helper data may be stored in a non-volatile memory of the device that includes the PUF.

(18) FIG. 3B illustrates the generation of helper data. In general, the helper data may be generated for the PUF corresponding to an SRAM as described with regards to FIG. 3A. As shown, an available code word basis **390** may be provided. The available code word basis **390** may include multiple code words that may be used in an error correction operation. A target data **391** may also be received and random numbers **392** may be generated. Based on the values of the bits of the target data **391** and the random numbers **392**, different code words from the available code word basis **390** may be selected to be used in helper data that is to be generated and used for the error correction operation. For example, the available code word basis **390** may include a code word for each available bit of the target data **391** and a code word for each available bit of the random numbers **392**. As shown, since the first bit of the target data **391** is a 0, then the code word C.sub.0 may not be used (e.g., a combination of the value of '0' and the code word is zero). Furthermore, since the next bit of the target data **391** is a '1,' then the code word C.sub.1 may be used. Based on

the values of the remaining bits of the target data **391**, the code word C.sub.6 may be used and the other code words may not be used (e.g., the seventh bit is a value of '1' and the other bits are at values of '0'). Furthermore, since the random numbers **392** include a value for the first, second, and fourth bits, then the code words d.sub.0, d.sub.1, and d.sub.3 may be used. Thus, the resulting code word basis **393** may include code words C.sub.1, C.sub.6, d.sub.0, d.sub.1, and d.sub.3. The resulting code word basis **393** may then be combined with an output of the PUF based on an XOR operation to generate the helper data.

(19) FIG. **4A** is a flow diagram of an example method **400** to determine a target data by using a physically unclonable function. The method **400** may be performed by processing logic that may include hardware (e.g., processing device, circuitry, dedicated logic, programmable logic, microcode, hardware of a device, integrated circuit, etc.), software (e.g., instructions run or executed on a processing device), or a combination thereof. In some embodiments, the method **400** may be performed by the physically unclonable function **110** and/or other device components **130** of the device **100** of FIG. **1**.

(20) As shown in FIG. **4A**, the method **400** may begin, at block **410**, with processing logic receiving an output of a physically unclonable function that corresponds, in this example, to an SRAM. For example, a device component may provide a request for the target data. In some embodiments, the target data may correspond to a cryptographic key to be used by the device component or a value that is used to generate the cryptographic key. The target data may correspond to any other type of secret data to be used by the device component. The processing logic may further receive helper data for the PUF (block **420**). For example, the helper data may be retrieved from a non-volatile memory. The processing logic may further perform an XOR operation between the output of the PUF and the helper data for the PUF (block **430**). For example, the values corresponding to the physical variations in the PUF and the helper data may be combined to generate a result. Furthermore, the processing logic may perform an error correction on the result of the XOR operation to identify code words (block **440**). Since the helper data may be a combination of the code words and a prior output of the PUF when the code words were generated, the result of the XOR operation between the helper data and the output of the PUF may correspond to a combination of the helper data, the prior output of the PUF, and the current output of the PUF. An XOR combination of the prior output of the PUF and the current output of the PUF may be an error term. For example, the error correction operation may remove the error term from the result of the XOR operation (e.g., the combination of the current output of the PUF and the prior output of the PUF) so that result of the error correction operation is the code words. Subsequently, the processing logic may determine the target data based on the identified code words (block **450**). The target data may be generated based on which code words have been used. For example, the code words may be decoded to retrieve the original random values that were encoded into the code words. The target data may then be retrieved by reading the first bits (e.g., the first group of bits) of the decoded values which may represent the bits of the original random value that were substituted or replaced by the bits of the target data.

(21) As such, the target data may be generated by performing an error correction operation for a combination of the current output of the PUF and the helper data. The use of the error correction operation may remove an error term from the combination of the current output of the PUF and the helper data to generate the code words that were generated when the target data was received to generate the helper data at a prior time. Subsequently, the code words may be used to determine the target data.

(22) FIG. **4B** illustrates the generation of the target data. In general, the target data may be generated by using the PUF corresponding to an SRAM as described with regards to FIG. **4A**. As shown, the PUF **460** may generate a current PUF output **461**. Furthermore, the current PUF output **461** and the helper data **462** (e.g., the helper data based on code words and a prior PUF output as described with respect to FIGS. **3A** and **3B**) may be combined based on an XOR operation to

generate a result data **464**. In some embodiments, the result data **464** may be a value that corresponds to an error term (e.g., the prior PUF output and the current PUF output) and the code words that were previously selected or generated based on the values of the target data and the random numbers. For example, the result data **464** may be an XOR combination of the current PUF output **461**, code words, and the prior PUF output when the code words were selected. The XOR combination of the current PUF output **461** and the prior PUF output may correspond to an error term. The error correction operation **465** may be performed on the result data **464** to remove the error term (e.g., the XOR combination of the current PUF output **461** and the prior PUF output) so that the code words **466** are remaining. Subsequently, the target data **467** may be derived from the code words **466**.

(23) In some embodiments, the physical variation of the PUF may correspond to differences in a resistance between a pair of nominally identical resistors. For example, a value of '0' may be assigned to a pair of resistors when the difference in resistance between the first and second resistor (i.e., due to random variation incurred during manufacturing) is larger than a predetermined threshold value and a value of '1' may be assigned to the pair of resistors when the difference in resistance between the first and second resistor is smaller than a predetermined threshold value. In some embodiments, if the difference in resistance between the pair of resistors does not exceed either of the threshold values that correspond to the values of '0' and '1' then the particular pair of resistors may not be used in the determination of the value of the PUF. In such an example, the helper data may identify pairs of resistors where the difference in resistance of the corresponding pairs is sufficiently large to be usable to determine a value of '0' or '1' from the pairs of resistors. The helper data may further identify an ordering of the values associated with the pairs of resistors that are used to determine the target data. In the same or alternative embodiments, the physical variation of the PUF may correspond to a race condition between two signal paths. For example, a value of '0' may be assigned if a first path satisfies the race condition before the second path and a value of '1' may be assigned if the second path satisfies the race condition before the first path. In such an example, the helper data may identify pairs of the paths that may be used to generate the target data.

(24) In some embodiments, the helper data may be a value that represents a combination between a first output value of the PUF and a code word. The code word may be generated by using bits of the target data. In some embodiments, the value may be an exclusive-or (XOR) combination of the first output value of the PUF and the code word. The code word may then be recovered by using the helper data with an output of a second fuzzy reading (i.e., a reading which includes some bits that are unstable and can vary between subsequent readings) of the PUF and then performing an error correction on the result to remove the error term. The target data may then be recovered from the resulting code word.

(25) FIG. 5 illustrates an example of a computer system **500** within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, may be executed. In alternative implementations, the machine may be connected (e.g., networked) to other machines in a LAN, an intranet, an extranet, and/or the Internet. The machine may operate in the capacity of a server or a client machine in client-server network environment, as a peer machine in a peer-to-peer (or distributed) network environment, or as a server or a client machine in a cloud computing infrastructure or environment.

(26) The machine may be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, a switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single machine is illustrated, the term "machine" shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

(27) The example computer system **500** includes a processing device **502**, a main memory **504** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), etc.), a static memory **506** (e.g., flash memory, static random access memory (SRAM), etc.), and a data storage device **518**, which communicate with each other via a bus **530**.

(28) Processing device **502** represents one or more general-purpose processing devices such as a microprocessor, a central processing unit, or the like. More particularly, the processing device may be complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processing device **502** may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device **502** is configured to execute instructions **526** for performing the operations and steps discussed herein.

(29) The computer system **500** may further include a network interface device **508** to communicate over the network **520**. The computer system **500** also may include a video display unit **510** (e.g., a liquid crystal display (LCD) or a cathode ray tube (CRT)), an alphanumeric input device **512** (e.g., a keyboard), a cursor control device **614** (e.g., a mouse), a graphics processing unit **522**, a signal generation device **516** (e.g., a speaker), graphics processing unit **522**, video processing unit **528**, and audio processing unit **532**.

(30) The data storage device **518** may include a machine-readable storage medium **524** (also known as a computer-readable medium) on which is stored one or more sets of instructions or software **526** embodying any one or more of the methodologies or functions described herein. The instructions **526** may also reside, completely or at least partially, within the main memory **504** and/or within the processing device **502** during execution thereof by the computer system **500**, the main memory **504** and the processing device **502** also constituting machine-readable storage media.

(31) In one implementation, the instructions **526** include instructions to implement functionality as described herein. While the machine-readable storage medium **524** is shown in an example implementation to be a single medium, the term “machine-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The term “machine-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of the present disclosure. The term “machine-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, optical media and magnetic media.

(32) Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

(33) It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the above discussion, it is appreciated that throughout the description, the present disclosure may refer to the action and

processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage devices.

(34) The present disclosure also relates to an apparatus for performing the operations herein. This apparatus may be specially constructed for the intended purposes, or it may comprise a general purpose computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a computer readable storage medium, such as, but not limited to, any type of disk including floppy disks, optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

(35) The algorithms and displays presented herein are not inherently related to any particular computer or other apparatus. Various general purpose systems may be used with programs in accordance with the teachings herein, or it may prove convenient to construct a more specialized apparatus to perform the method. The structure for a variety of these systems will appear as set forth in the description below. In addition, the present disclosure is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the disclosure as described herein.

(36) The present disclosure may be provided as a computer program product, or software, that may include a machine-readable medium having stored thereon instructions, which may be used to program a computer system (or other electronic devices) to perform a process according to the present disclosure. A machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium such as a read only memory ("ROM"), random access memory ("RAM"), magnetic disk storage media, optical storage media, flash memory devices, etc.

(37) In the foregoing disclosure, implementations of the disclosure have been described with reference to specific example implementations thereof. It will be evident that various modifications may be made thereto without departing from the broader spirit and scope of implementations of the disclosure as set forth in the following claims. The disclosure and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

1. A method comprising: providing an input target data to a physically unclonable function (PUF), wherein an output of the PUF is associated with a physical variation of a device; generating helper data associated with the physical variation based on the output of the PUF and the input target data; storing the helper data at a memory of the device; receiving a request to generate a target data with the PUF; providing the helper data stored at the memory to the PUF; and generating the target data based on the output of the PUF and the helper data, wherein the target data matches the input target data.

2. The method of claim 1, further comprising: performing an error correction operation on a combination of the helper data and the output of the PUF; and identifying one or more code words associated with the target data responsive to the error correction operation.

3. The method of claim 2, further comprising generating the target data based on a combination of the one or more code words and the output of the PUF.

4. The method of claim 2, further comprising receiving a random value from a random number generator, wherein the one or more code words associated with the error correction operation is generated based on the random value.

5. The method of claim 2, further comprising generating the helper data based on an exclusive-or (XOR) operation between the output of the PUF and the one or more code words.
6. The method of claim 1, further comprising: identifying, by the helper data, one or more bits associated with the physical variation; and replicating the input target data as an output value of the PUF based on the one or more bits.
7. The method of claim 1, further comprising retrieving the helper data from the memory at a subsequent instance to generate the target data.
8. The method of claim 1, further comprising receiving the request to generate the target data with the PUF subsequent to generating and storing the helper data at the memory.
9. The method of claim 1, wherein the request to generate the target data is responsive to a request from a device component for a cryptographic key corresponding to the target data.
10. The method of claim 1, further comprising performing an exclusive-or (XOR) operation on the helper data and the output of the PUF to produce a result data corresponding to a combination of the helper data, a prior output of the PUF, and the output of the PUF.
11. The method of claim 1, wherein the PUF corresponds to a static random-access memory (SRAM).
12. A system comprising: a memory; and a processing device, operatively coupled with the memory to: provide an input target data to a physically unclonable function (PUF), wherein an output of the PUF is associated with a physical variation of a device; generate helper data associated with the physical variation based on the output of the PUF and the input target data; store the helper data at a memory of the device; receive a request to generate a target data with the PUF; provide the helper data stored at the memory to the PUF; and generate the target data based on the output of the PUF and the helper data, wherein the target data matches the input target data.
13. The system of claim 12, wherein an error correction operation is performed on a combination of the helper data and the output of the PUF to identify one or more code words associated with the target data.
14. The system of claim 12, wherein the processing device is further to: identify, by the helper data, one or more bits associated with the physical variation; and replicate the input target data as an output value of the PUF based on the one or more bits.
15. The system of claim 12, wherein the helper data is retrieved from the memory at a subsequent instance to generate the target data.
16. The system of claim 12, wherein the request to generate the target data is responsive to a request from a device component for a cryptographic key corresponding to the target data.
17. The system of claim 12, wherein the PUF corresponds to a static random-access memory (SRAM).
18. A non-transitory computer readable medium comprising instructions that, when executed by a processing device, cause the processing device to perform operations comprising: providing an input target data to a physically unclonable function (PUF), wherein an output of the PUF is associated with a physical variation of a device; generating helper data associated with the physical variation based on the output of the PUF and the input target data; storing the helper data at a memory of the device; receiving a request to generate a target data with the PUF; providing the helper data stored at the memory to the PUF; and generating the target data based on the output of the PUF and the helper data, wherein the target data matches the input target data.
19. The non-transitory computer readable medium of claim 18, the processing device to perform operations further comprising: performing an error correction operation on a combination of the helper data and the output of the PUF; and identifying one or more code words associated with the target data responsive to the error correction operation.
20. The non-transitory computer readable medium of claim 18, the processing device to perform operations further comprising: identifying, by the helper data, one or more bits associated with the

physical variation; and replicating the input target data as an output value of the PUF based on the one or more bits.
